



6th Summer School in Social Statistics

Complex problems & complex models

Stefano Campostrini



Università
Ca' Foscari
Venezia





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Governance
& Social Innovation

Complexity (cum plectere)



It is not
complication



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First step in your research



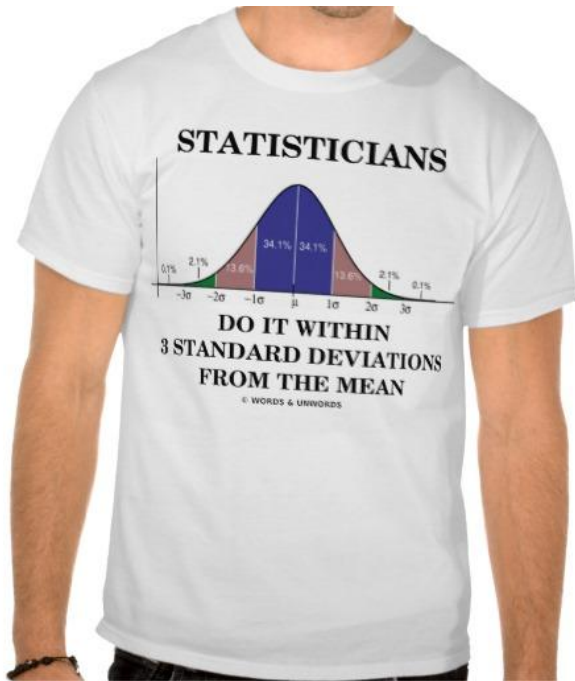
Ask yourself why you are doing it!!!



First step in your research

A brief introduction to the very basic of EPISTEMOLOGY

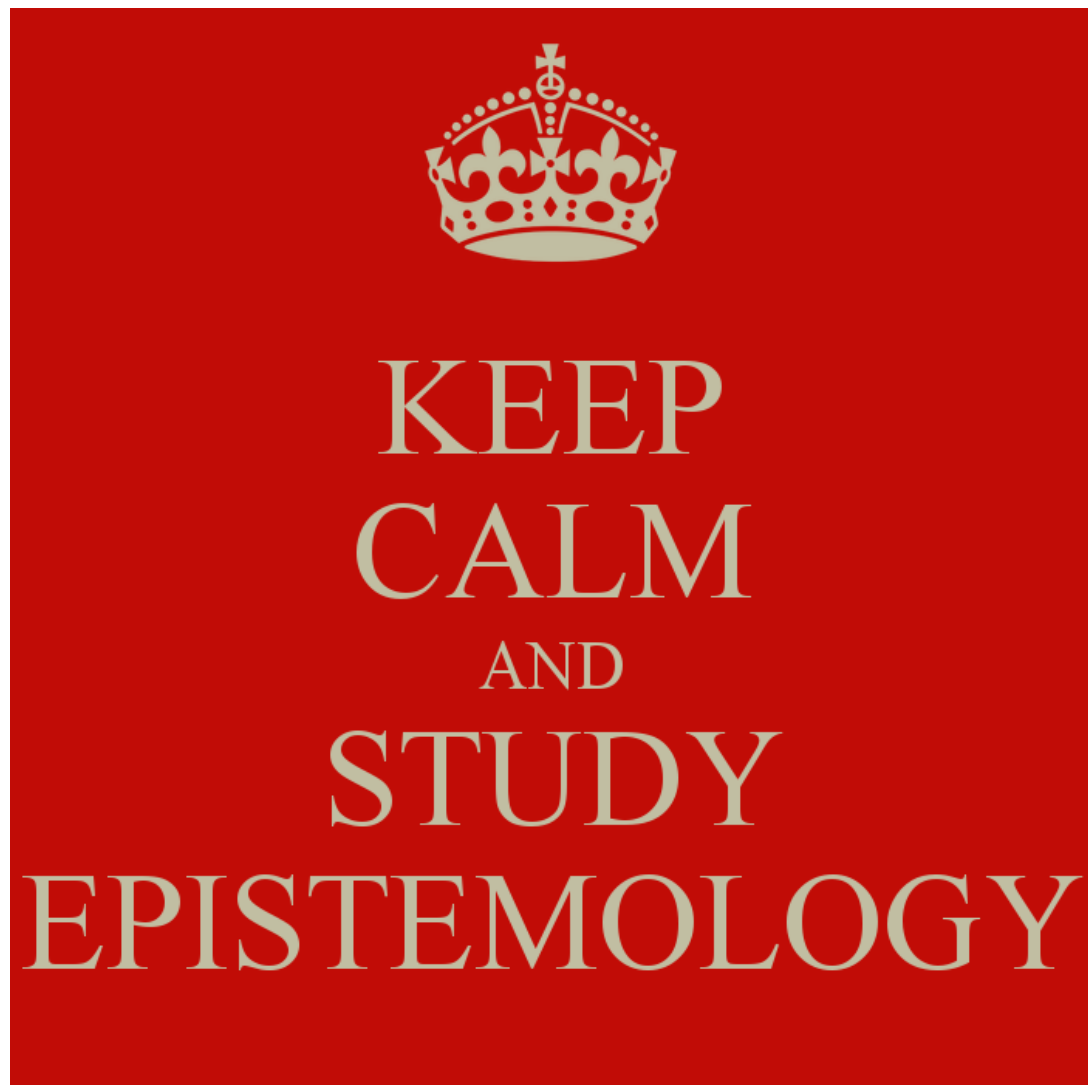
for Statisticians



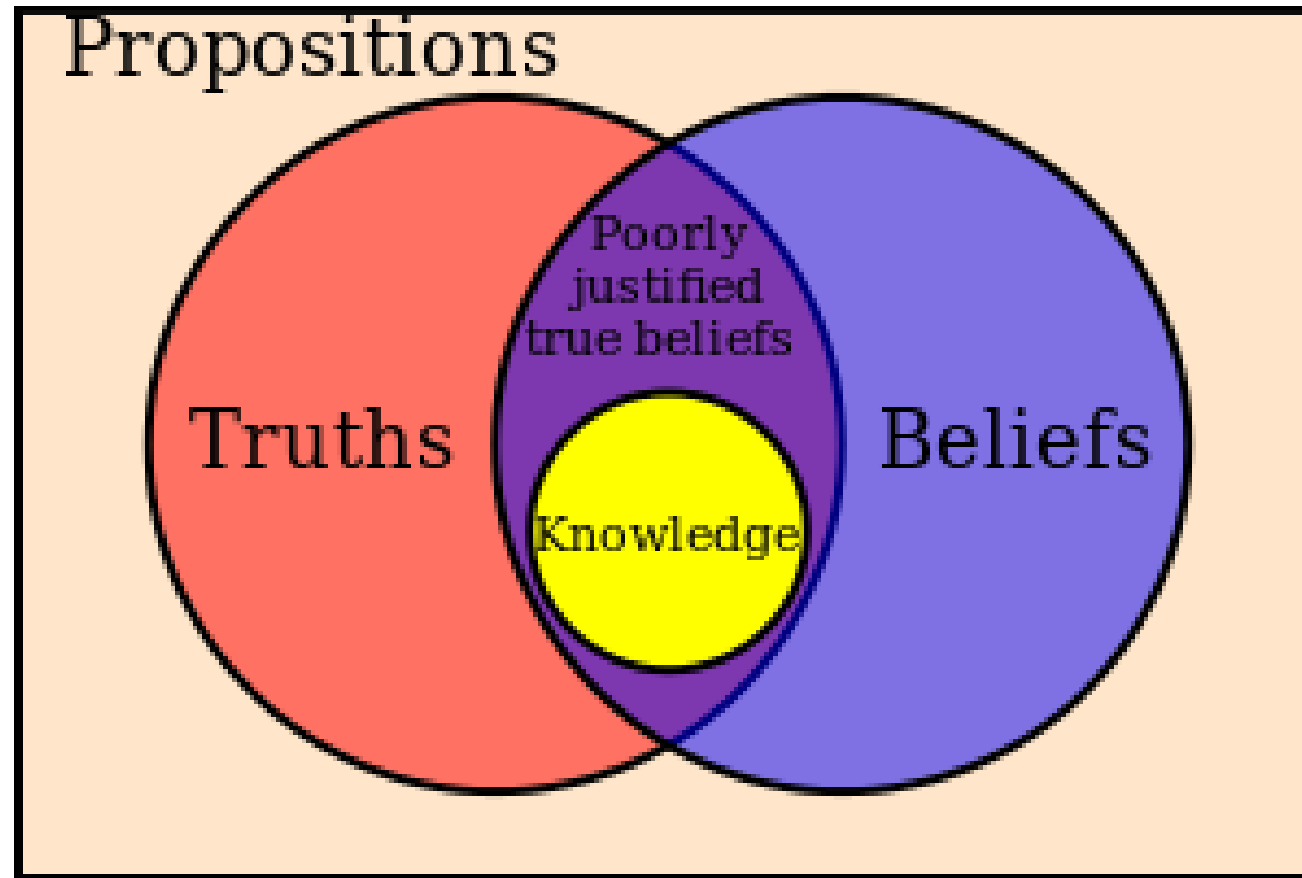


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First step in your research

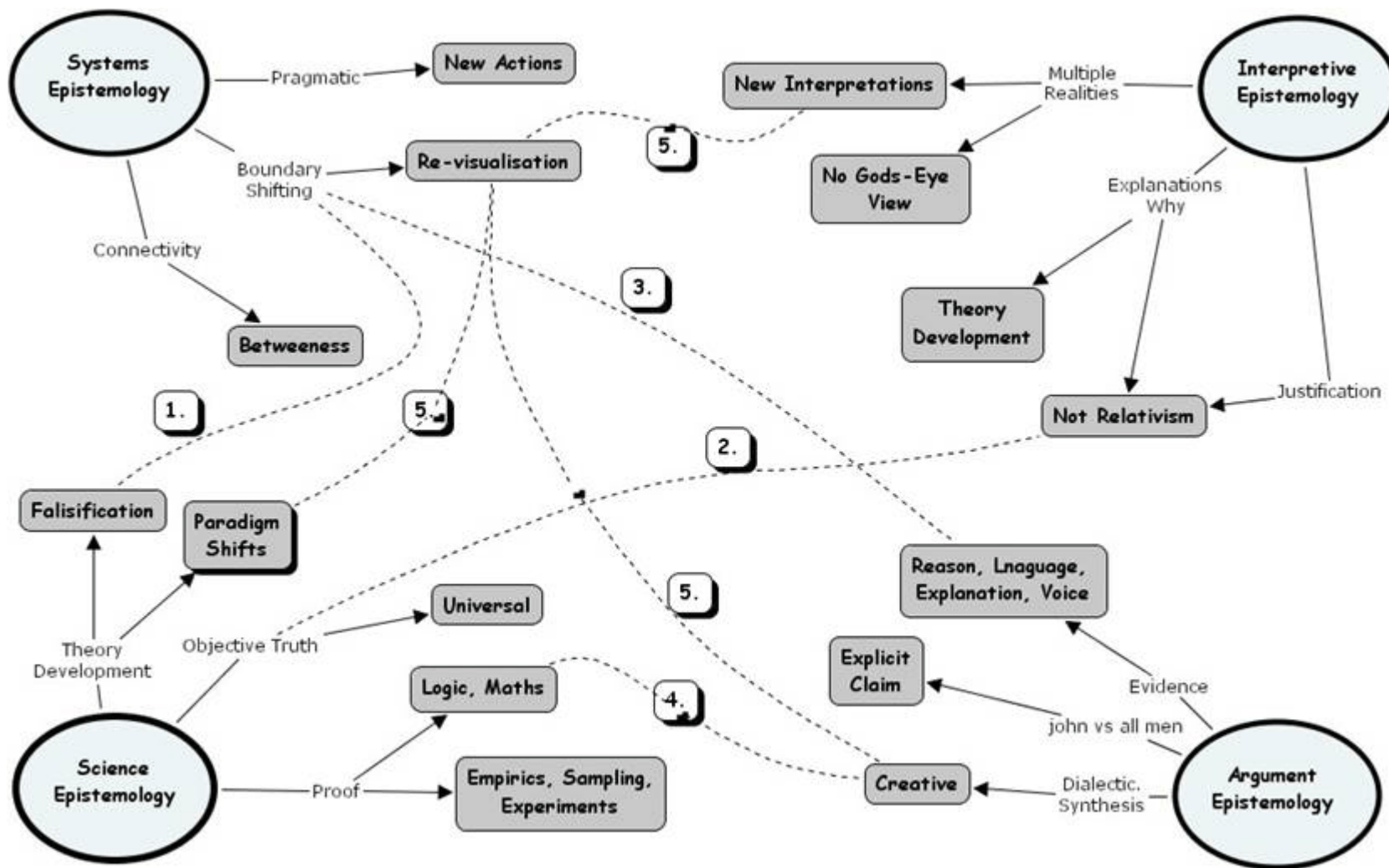


It is (almost) all in this picture



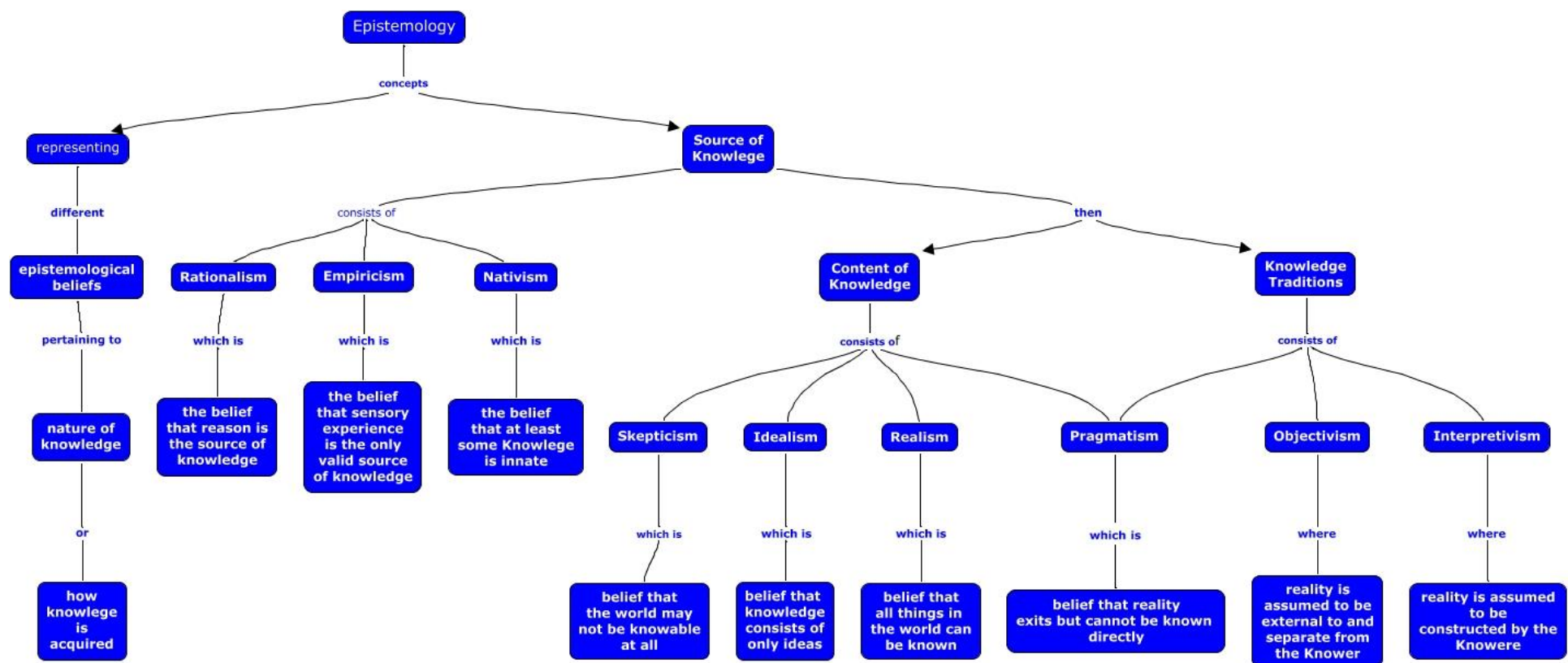


Or in this one





This is perhaps better

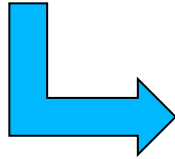




Scientific approach

Aristotele (383–322 bc)

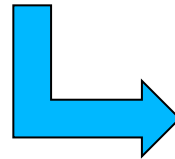
Induction as base for knowledge of superior elements



Galileo (1564-1642)

Law of nature are mathematical

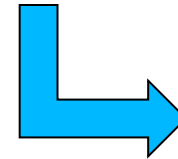
Inductive reasoning confirms mathematical law



Leibniz (1646-1716)

Newton (1642-1727)

Models can
explain the
world



Popper (1902-1994)

Falsificationism



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Statistical Epistemology

How not to lie
with Statistics

Hume (1711-1776)

Induction

Kant (1724-1804)

The method has a
subjective component,
still is not relative

*Statisticians must be clear about
the epistemic goals of their methods*



Statistical Epistemology

Epistemic
goals

Null hypothesis
vs Models

With a statistical test we are
not saying what is the
probability of something to
happen, but simply how
frequent it is

Inductive - inference

All models are wrong, but
some are useful (Box)

Bayes (1702-1761)

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Prior probability
(initial degree of belief in A)

Conditional probability (likelihood)
(degree of belief in B, given A true)

Bayes' Theorem: epistemological interpretation

$P(\text{parameters} | \text{data}) = P(\text{data} | \text{parameters})P(\text{parameters})$

Knowledge = likelihood * prior



Statistical Epistemology

One major
concept:
significance

Statistical
significant

Evidence based on the data
Beside any possible random
(sampling) error effect

Necessary but
not sufficient
condition

...

Significant

Evidence based on a theory
behind the data
(e.g. clinically significant =
relevant from
epidemiological/clinical
experience)



Plato's Republic, VII,7

Behold! human beings living in an underground den, which has a mouth open towards the light and reaching all along the den; here they have been from their childhood

[...]

Above and behind them a fire is blazing at a distance [] you will see if you look a low wall distance, [...] you will see, if you look, a low wall built along the way, like the screen which marionette players have in front of them, over which they show the puppets.

[] ...

They see only their own shadows, or the shadows of one another, which the fire throws on the opposite wall of the cave [...]

To them, I said, the truth would be literally nothing but the shadows of the images.



Public Health Surveillance

World Health Organization defines *public health surveillance* as

“the continuous, systematic collection, analysis and interpretation of health-related data needed for the planning, implementation, and evaluation of public health practice. Such surveillance can:
serve as an early warning system for impending public health emergencies;
document the impact of an intervention, or track progress towards specified goals; and
monitor and clarify the epidemiology of health problems, to allow priorities to be set and to inform public health policy and strategies.”

http://www.who.int/topics/public_health_surveillance/en/

trends

trends and association

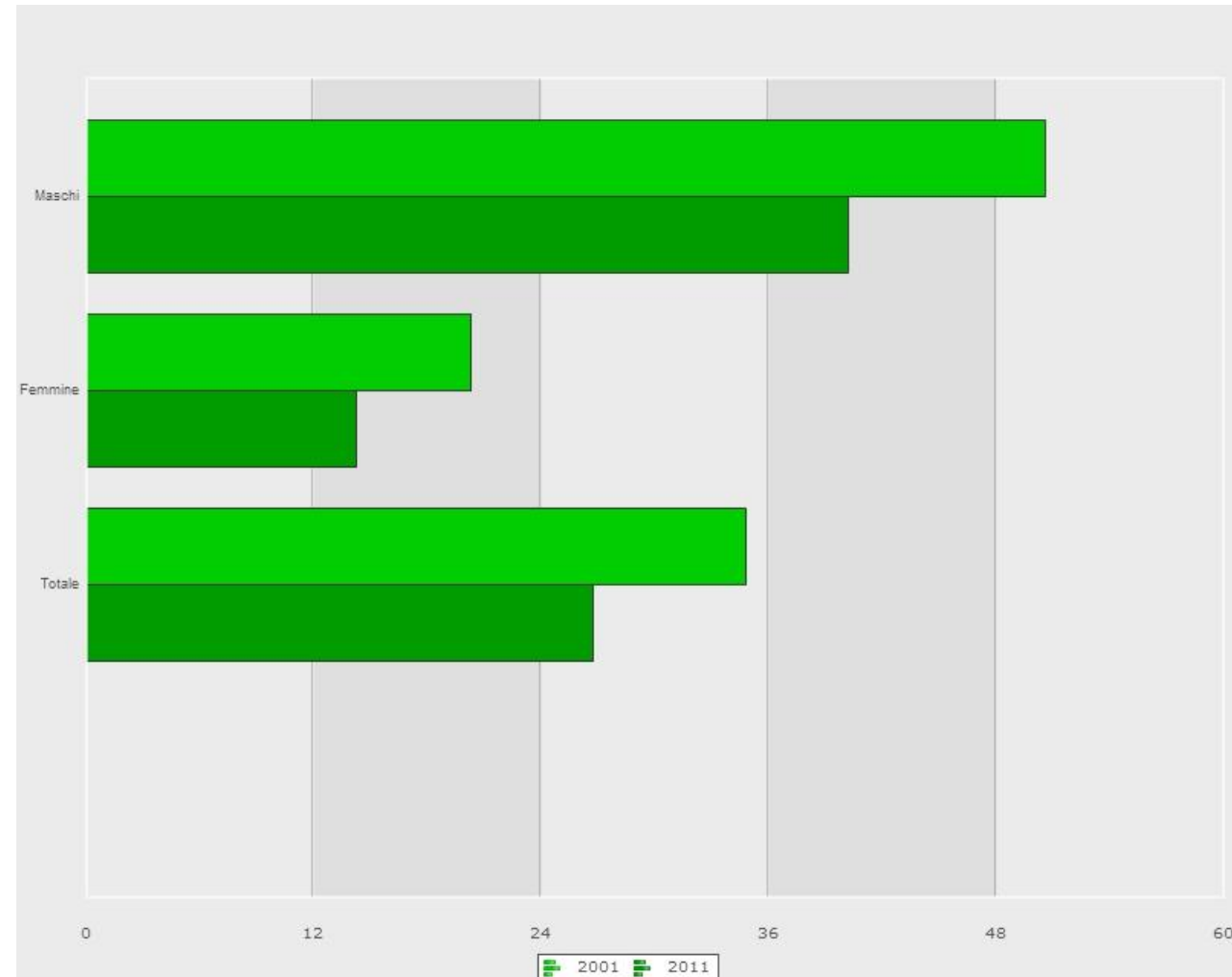
change analysis

evaluation

*BRF Surveillance produce,
from a statistical point of view,
a series of independent samples
on the same variables and
on (almost) the same population*

Some standard analysis...

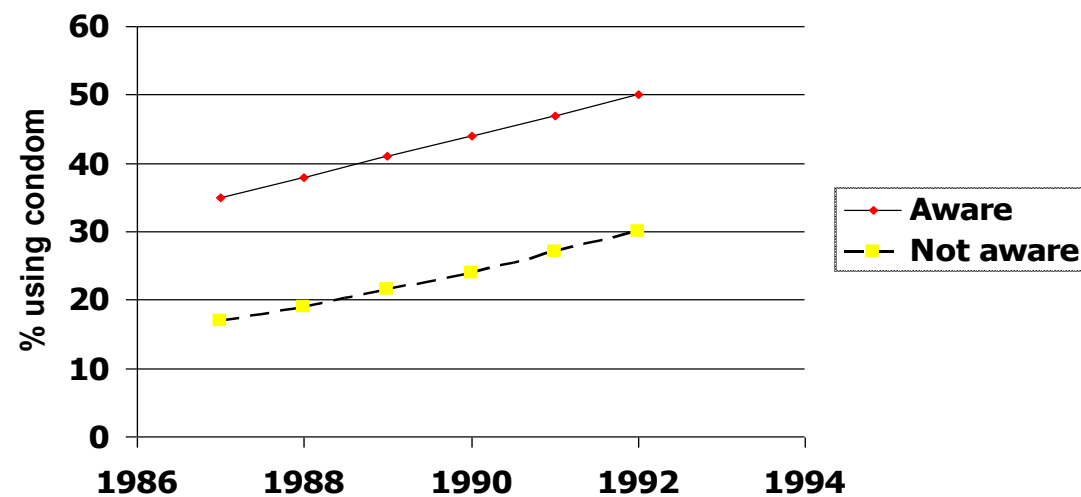
Persons that drink alcohol every day (%)





Condom Use and Awareness: *logit models*

LAH (Scottish) data 1987-92



Condom Use and Knowledge: *logit models*

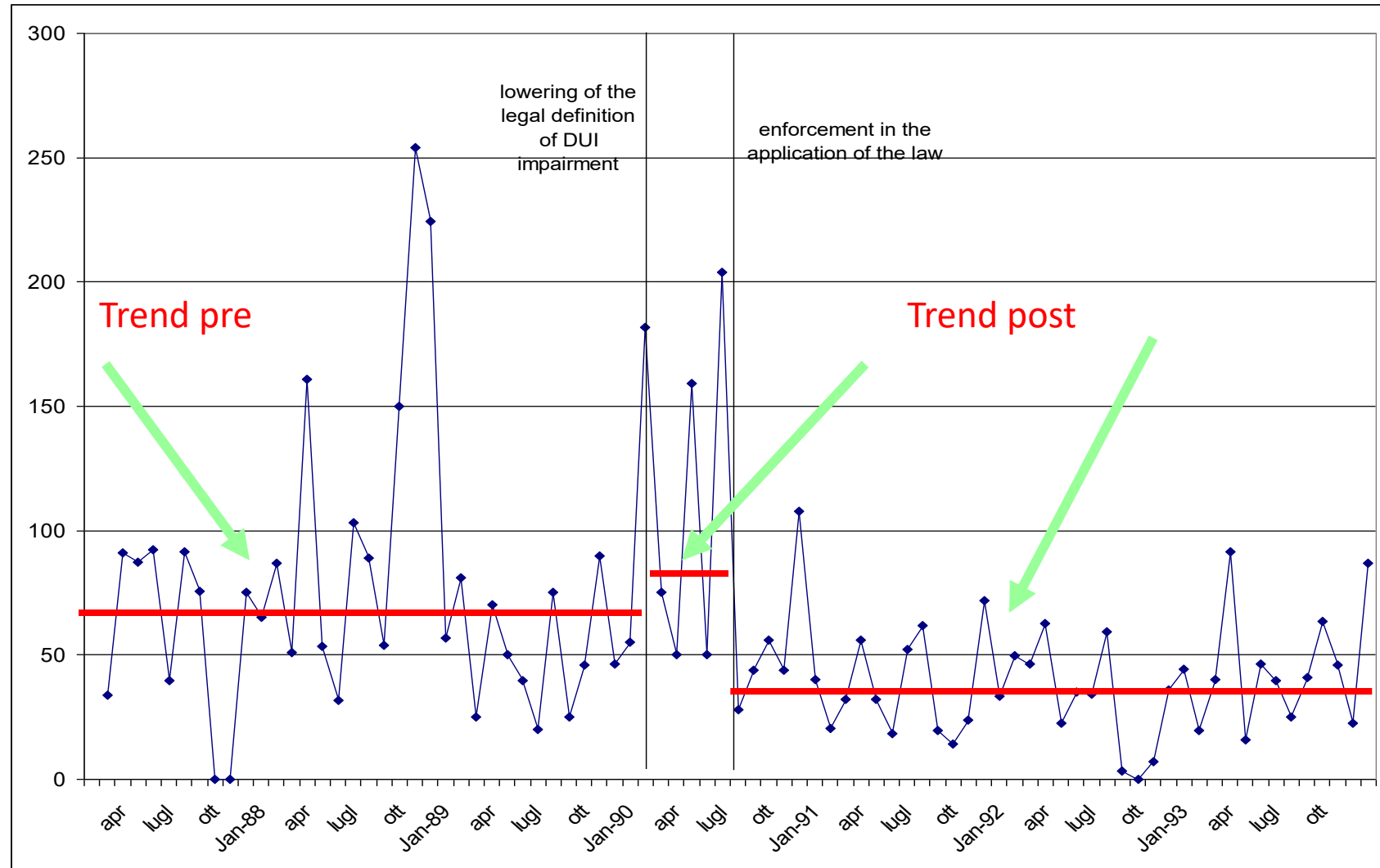
LAH (Scottish) data 1987-92

MODEL	df	G ²	P
(1) CAY (saturated)	-	-	1.00
(2) CA, AY, CY	4	3.22	.521
(3) CY, AY	5	369.07	.000



An increase in condom use has been observed; could this be attributable to changes in AIDS-related knowledge&att. ?

The time series: monthly average estimated number of episodes of drink and driving per 1,000 persons. BRFSS California data, n=20,006



Results

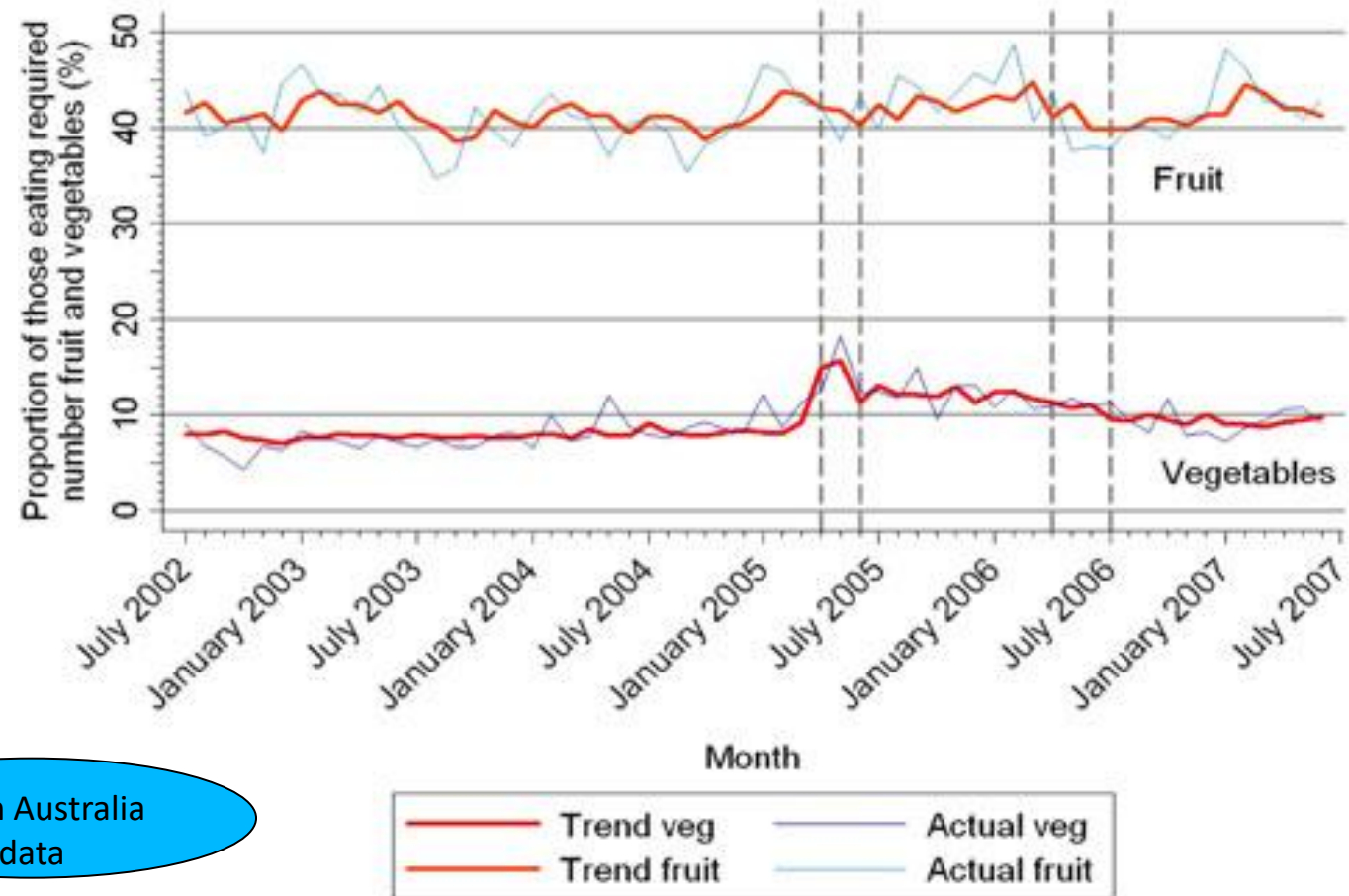
Table 1: The different ARIMA models

$Y_t = \omega_1 I_{t_1} + N_t$	I_{t_1} is equal to 1 from January 1990
$Y_t = \omega_2 I_{t_2} + N_t$	I_{t_2} is equal to 1 from July 1990.
$Y_t = N_t + \omega_1 I_{t_1} + \omega_2 I_{t_2}$	

Parameter	Value	Std. Error	T.Ratio	P-value
ω_1	0.03161	0.02464	1.28	0.204
ω_2	-0.07976	0.02429	-3.28	0.0015

Only the second intervention has had a significant (positive) impact

The use of chronic disease risk factor surveillance systems for evidence-based decision-making: nutrition as examples



South Australia
data

La prospettiva della compressione della morbidità...

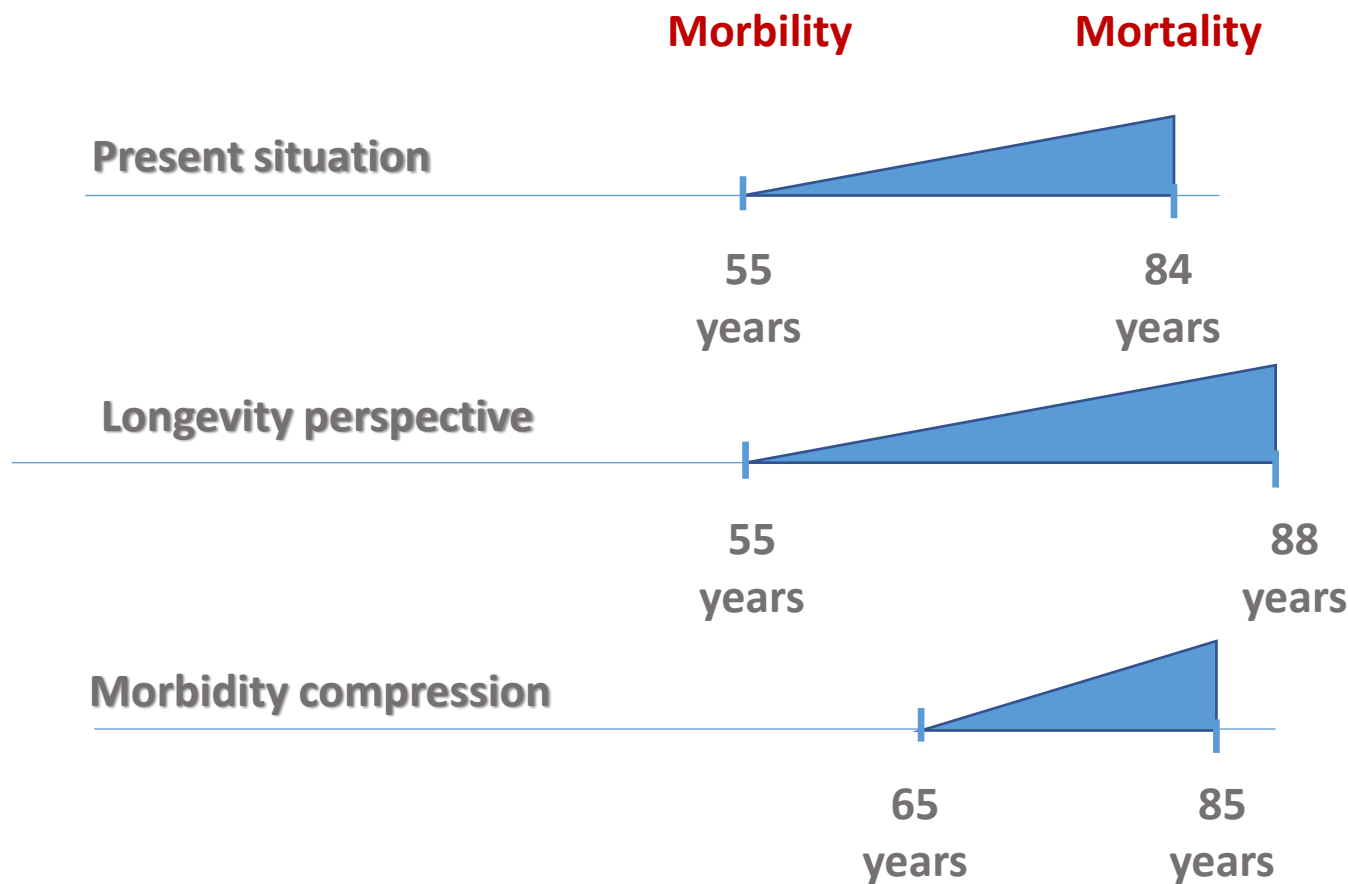
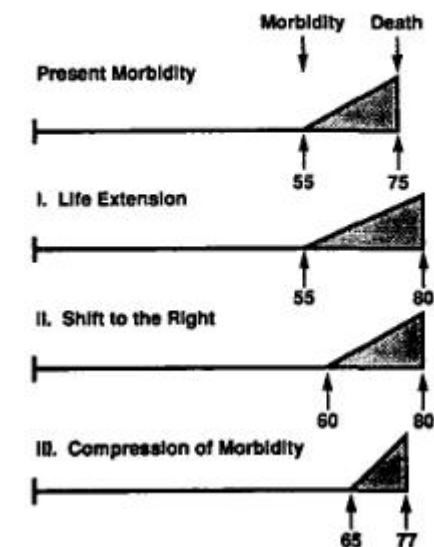


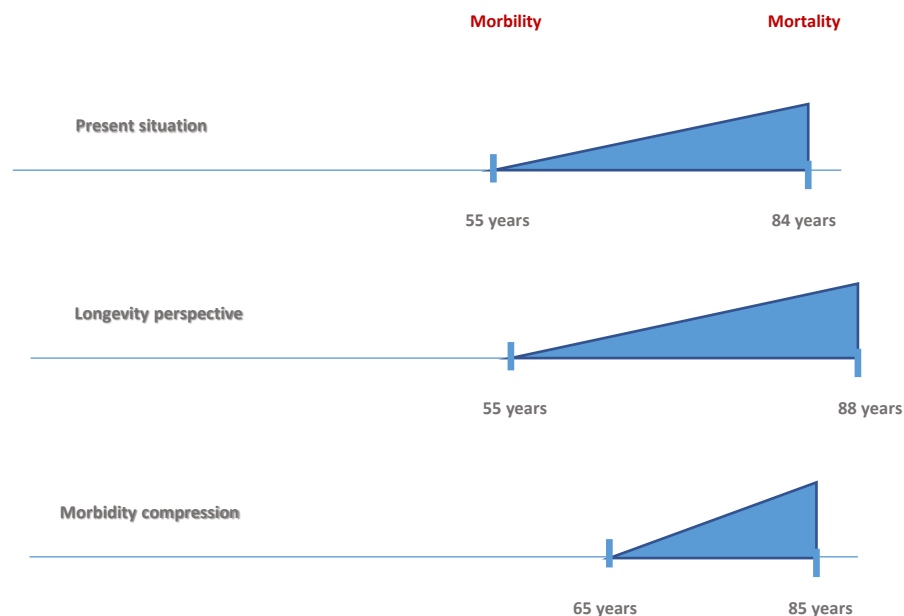
Figure 1
Scenarios for Future Morbidity and Longevity



Fries, J.F. (1980). "Aging, Natural Death, and the Compression of Morbidity." *New England Journal of Medicine*, 303(3), 130–135

Fries, J.F. (1990). "The Compression of Morbidity: Progress and Potential." *Journal of Insurance Medicine*, 22(2), 91–97.

Le rilevanti domande (tecniche) legate alla compressione della morbidità



1. Quando inizia la “cronicità”?
2. Come inizia ed evolve?
3. Fattori differenziali per sottogruppi di popolazione

I nostri dati: indagine PASSI

Sistema Italiano di sorveglianza di comportamenti e fattori di rischio



Istituto Superiore di Sanità
EpiCentro - L'epidemiologia per la sanità pubblica

Indice A-Z dei contenuti
Cerca..

Sorveglianza PASSI

Home | EpiCentro

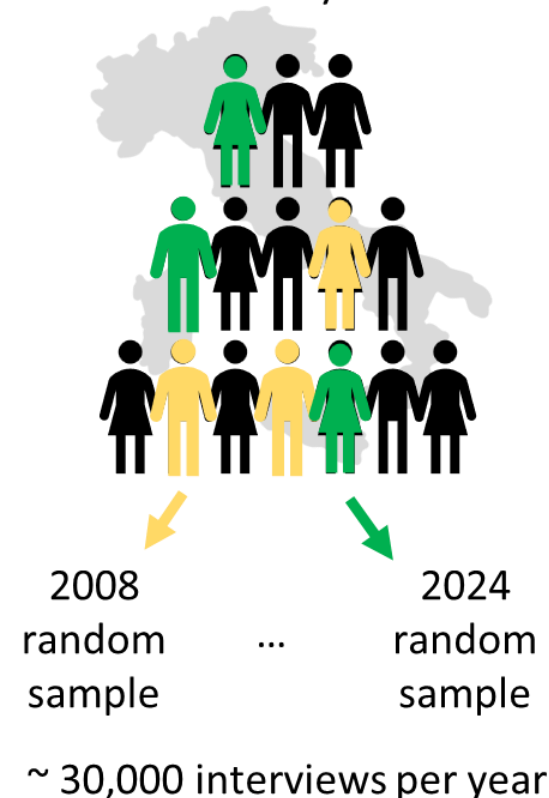
- PASSI**
- PASSI sur
- veillance
- edizione italiana

Progressi delle Aziende Sanitarie per la Salute in Italia (Passi)
The Italian behavioral risk factor surveillance system

Non-communicable chronic diseases cause the major part of the disease burden and have great economic impact, especially in countries with advanced economies. Behavioural risk factors play an important role in their development.

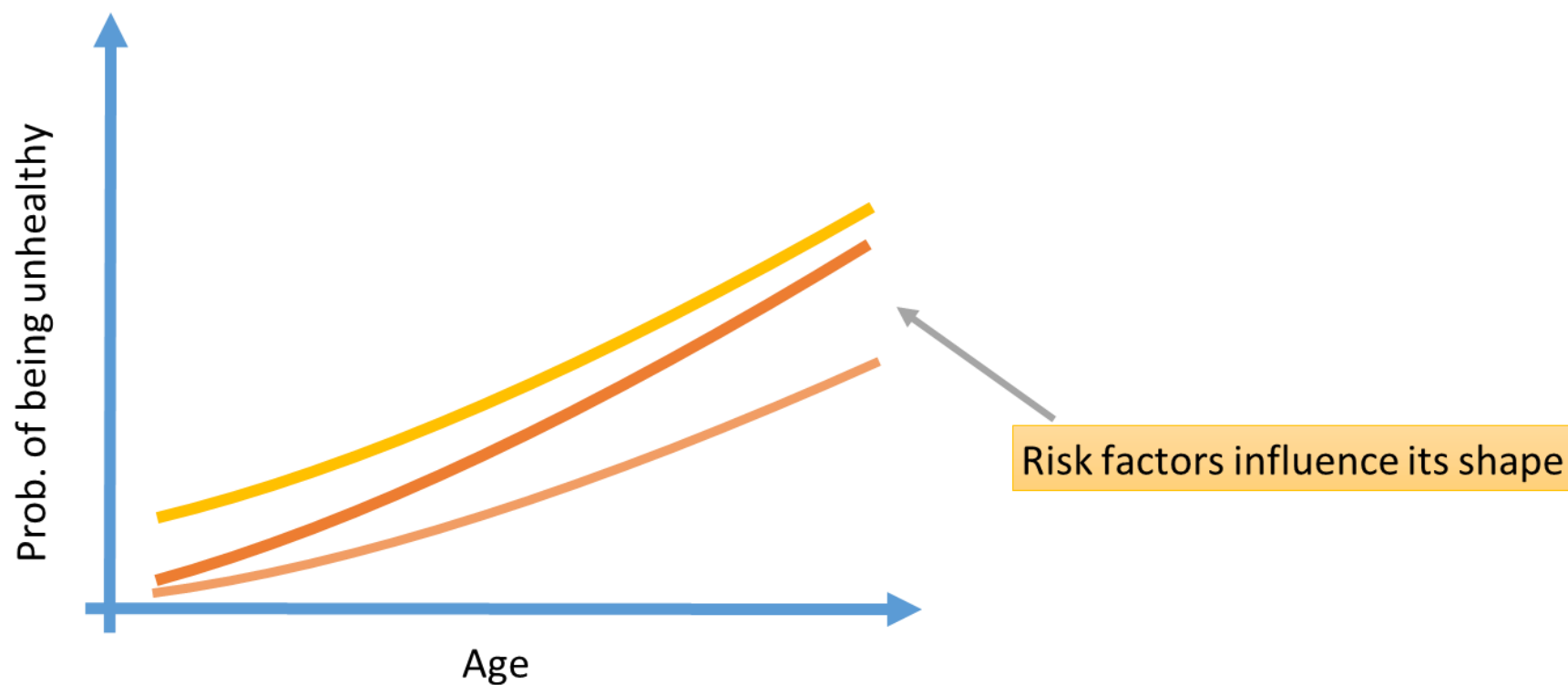
To plan, implement and evaluate preventive interventions on chronic diseases, timely information are needed not only on health status and behaviours of the population, but also on their knowledge and perceptions about healthy habits and lifestyle.

Cross-sectional repeated survey



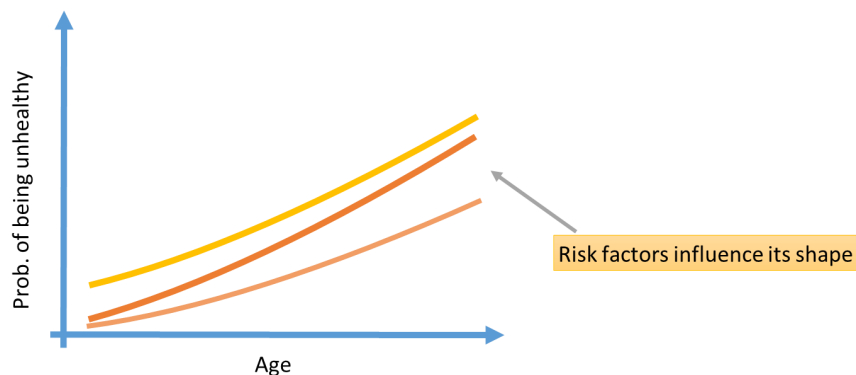
Morbidity curve

La curva di morbidità riflette la relazione tra età e probabilità di **non essere in salute** per un individuo in una certa popolazione



METODOLOGICAL NOTE

- **Repeated cross-sectional surveys**, particularly when conducted at **short and consecutive time intervals**, can provide evidence on the **evolution of phenomena over time**—a feature that is often highly relevant in public health, sometimes even more so than the mere estimation of prevalence.
- **However, they require “appropriate” models.**
- *Campostrini, S. (1992). Indagini socio-sanitarie continue. Problemi di misura e di analisi [Continuous health and social surveys: Measurement and analytical issues]. Doctoral dissertation in Statistics, University of Padua.*



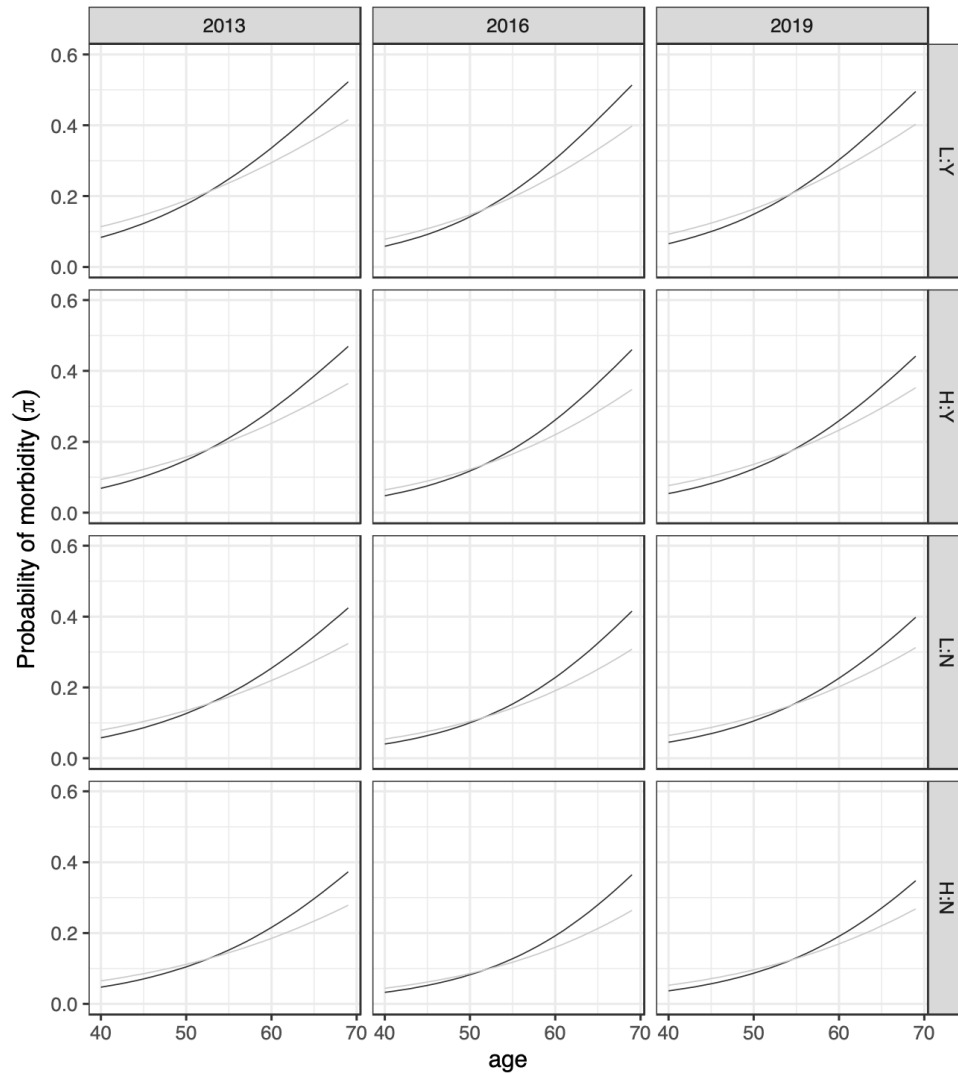
Lifestyle and Health, RUHBC – Edinburgh University
BRFSS – CDC Atlanta



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Studying morbidity compression

Unfolding the SEcrets of LongEvity: Current Trends and future prospects" (SELECT)



sex

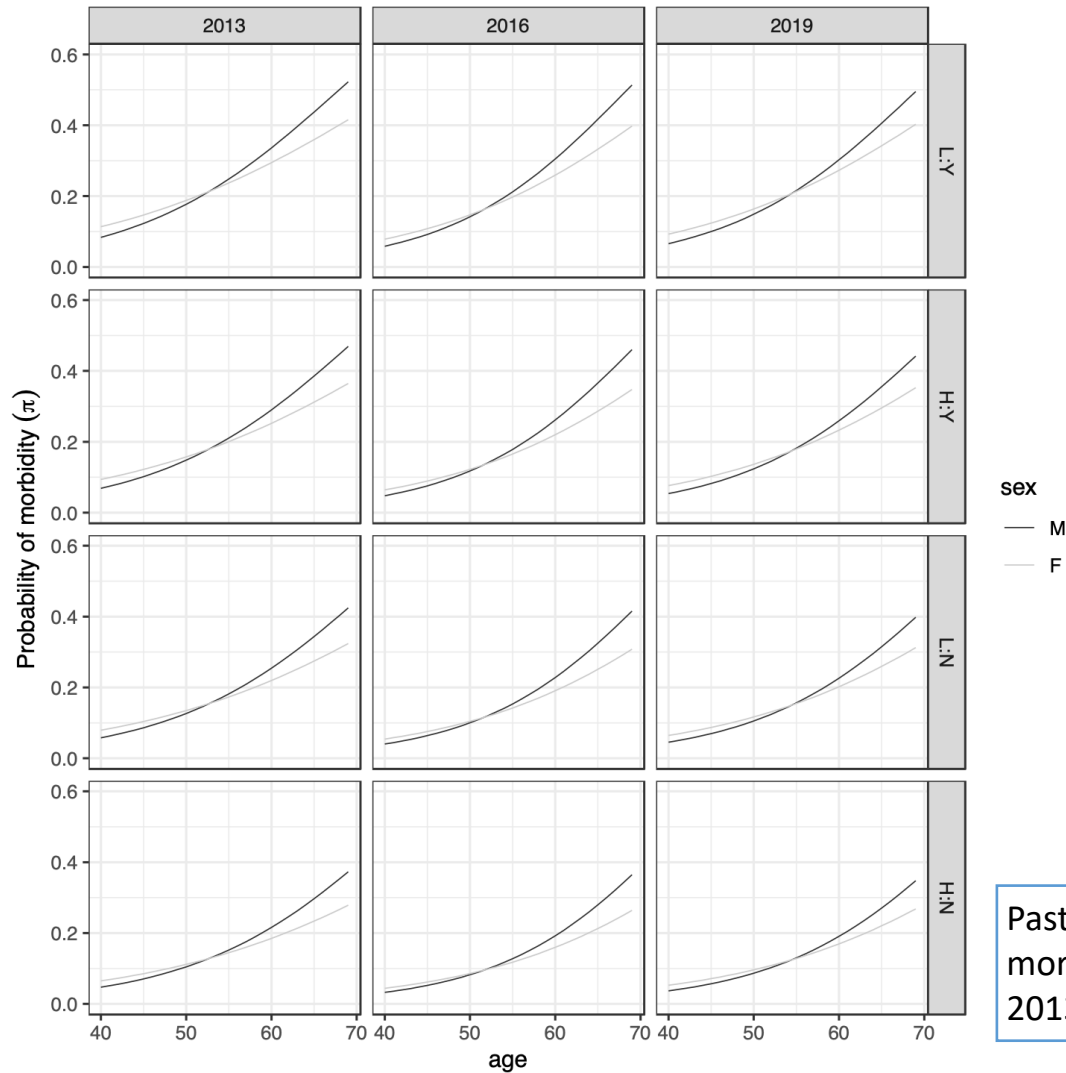
— M
- - F

Posterior means and 95% credible intervals for π_i by age, year, and other covariates. Panels in each column refer to a given year, while panels in each row refer to a given combination of levels of the other two covariates education and economic difficulties. The labels for the rows are the following: H:N means education = medium-high and economic difficulties = no; L:N means education = low and economic difficulties = no; H:Y means education = medium-high and economic difficulties = yes; L:Y means education = low and economic difficulties = yes.

Pastore A., Tonellato S., Aliverti E., Campostrini S., When does morbidity start? An analysis of changes in morbidity between 2013 and 2019 in Italy, *Statistical Methods and Application*, 2022



Indicatore di morbidità «grezzo»: presenza di almeno una patologia diagnosticata



Studying morbidity compression

Le curve si «abbassano»
negli anni, pur
mantenendo le
differenze tra generi e
classi SE

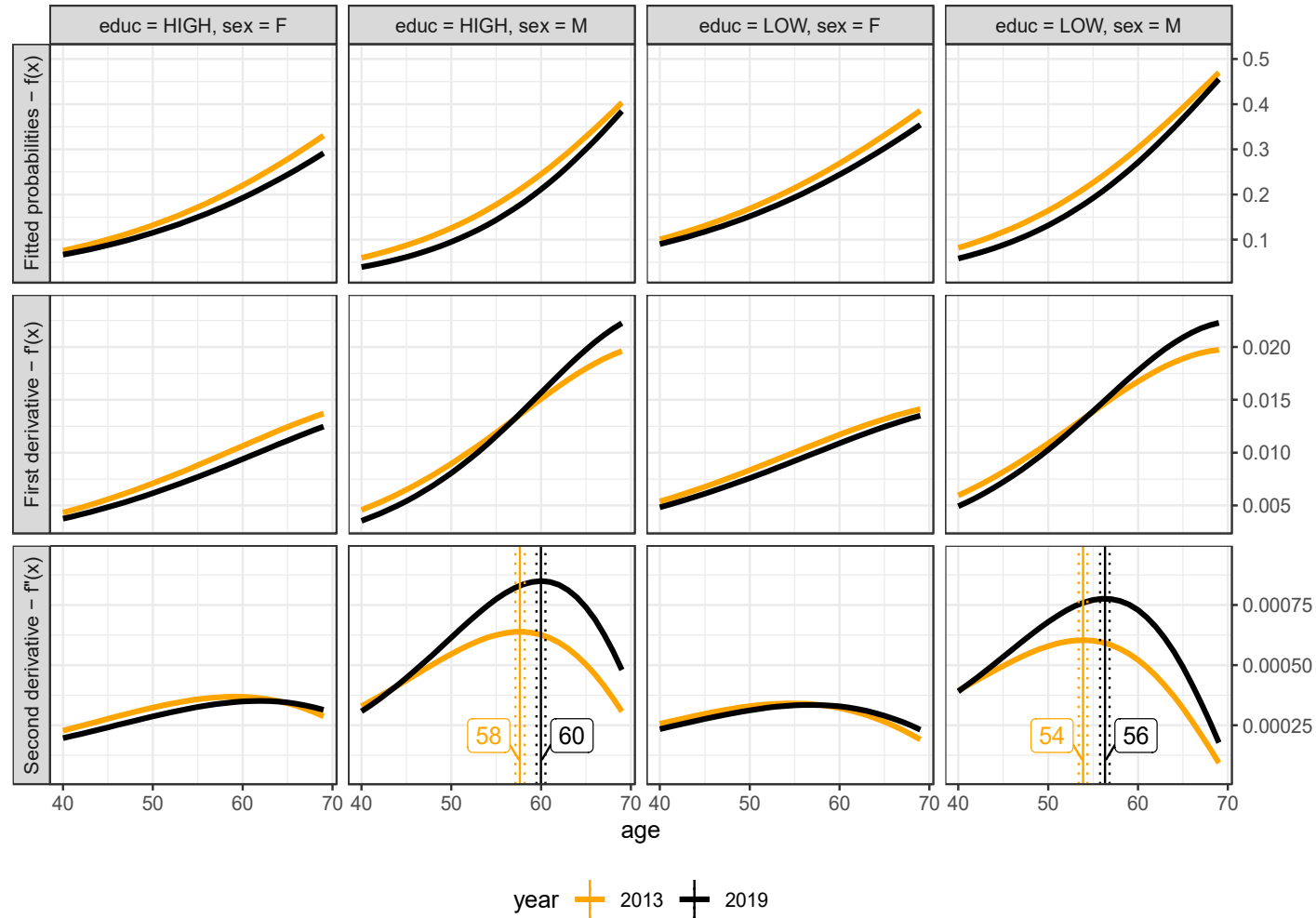
able intervals for π by age, year, and column refer to a given year, while combination of levels of the economic difficulties. The g: H:N means education = difficulties = no; L:N means education = no; H:Y means education = economic difficulties = yes; L:Y means education = low and economic difficulties = yes.

Pastore A., Tonellato S., Aliverti E., Campostrini S., When does morbidity start? An analysis of changes in morbidity between 2013 and 2019 in Italy, *Statistical Methods and Application*, 2022



Indicatore di morbidità «grezzo»: presenza di almeno una patologia diagnosticata

Studying morbidity compression (functional analysis)



Living longer and living healthier? Analysis of compression of morbidity in Italy (*to be published*)



Top panels: predicted probability across subgroups, as a function of age. Points denote the observed proportions of individuals with chronic condition. Middle and bottom panels represent the estimated first and derivatives of the function reported on top, respectively. Vertical lines denote the estimated maxima of the second derivatives. Posterior medians are represented with solid lines, while shades represent 95% credible intervals

Studying comorbidity across socio-economical populations considering the «weight» of each disease

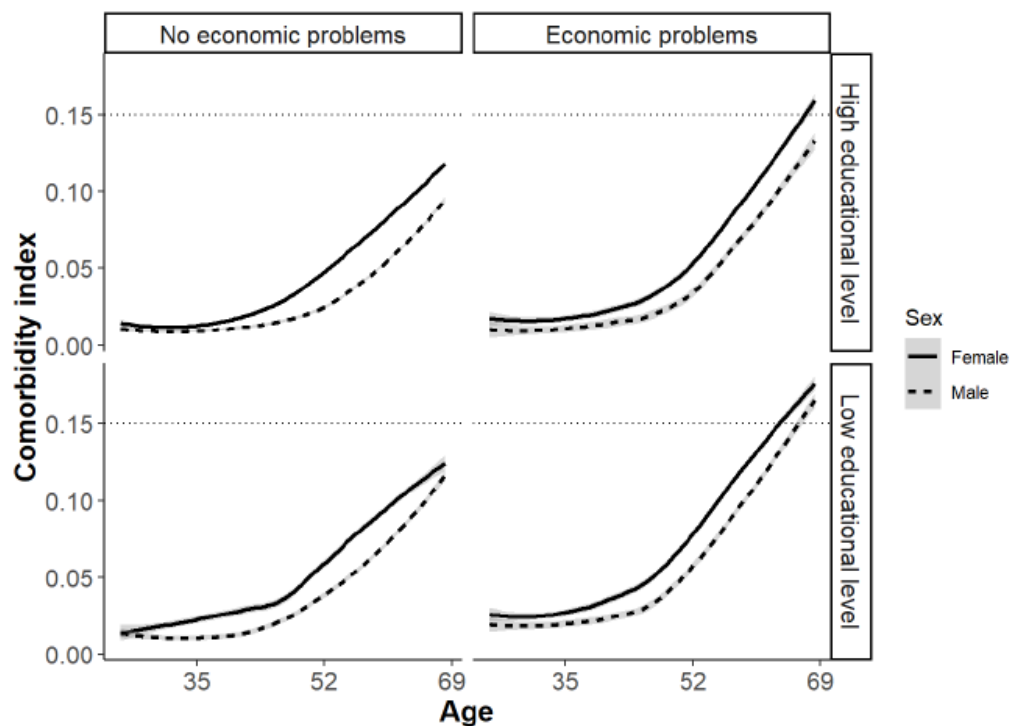


Figure 4 Predictions of the comorbidity index across age considering sub-populations characterized by different levels of education (low-high) and economic status (no economic problems-economic problems) in 2019. The grey area represents the prediction interval at level 0.95.



Andreella, A., Monasta, L., & Campostrini, S. (2025). Analysis of multimorbidity compression using a latent variable in a mixed mixture model. *Population Health Metrics*, 23(1), 58.

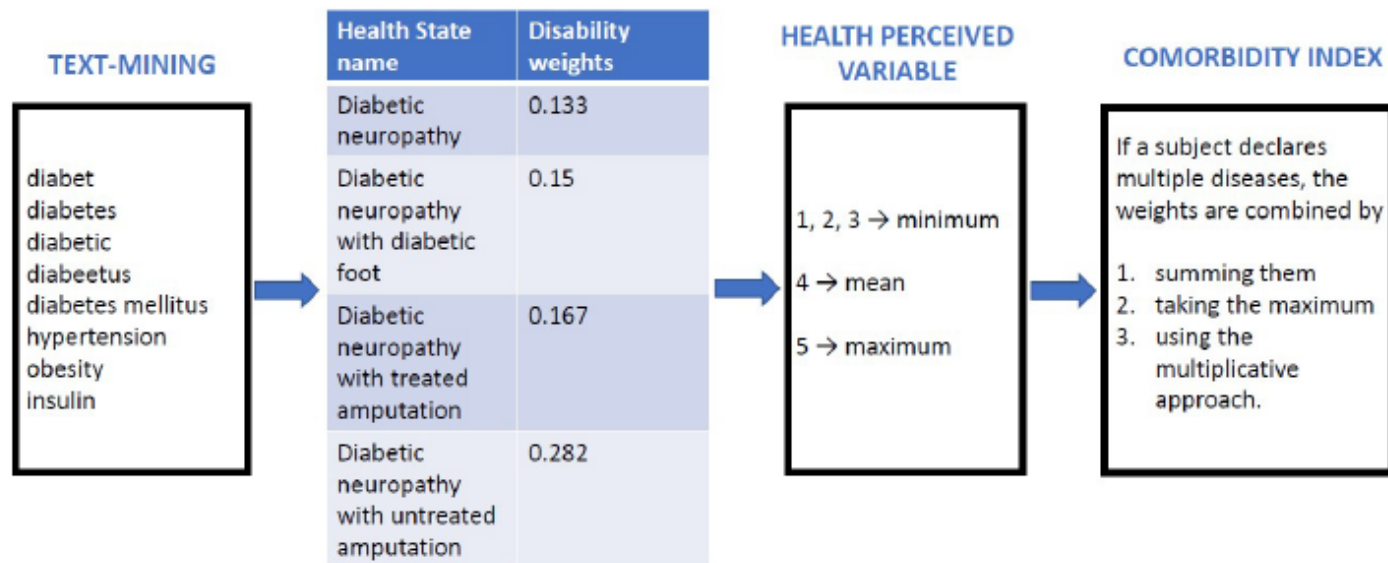


Figure 3 Steps to associate the weights coming from the GBD to the NCDs declared in the Italian surveillance system PASSI (year 2019).

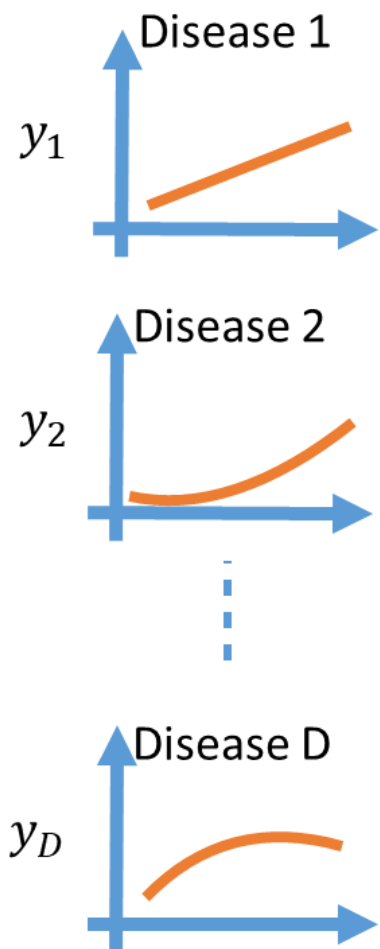
Comorbidità e pesi diversi alle patologie



Studying comorbidity across socio-economical populations

Andreella, A., Monasta, L., & Campostrini, S. (2025). Analysis of multimorbidity compression using a latent variable in a mixed mixture model. *Population Health Metrics*, 23(1), 58.

Informazione disponibile su cinque malattie croniche



$$\mathbf{y} = [y_1, y_2, \dots, y_D]$$

Approccio multivariato riconosce la natura multi-dimensionale del problemare e considera **ogni malattia “separatamente”**, essendo ognuna caratterizzata diversamente.

PRO:

Approccio multivariato permette l'**identificazione di caratteristiche specifiche** di ogni malattia ma garantendo la possibilità di studiare **anche la relazione tra malattie**.



My team: So.Sta.



Michele Marzulli



Stefano Campostrini



Stefano Tonellato



Andrea Pastore

Sociologists



Michele Bertani



Gaia Bertarelli



Mattia Stival



Marta Pittavino

Statisticians



Angela Andreella



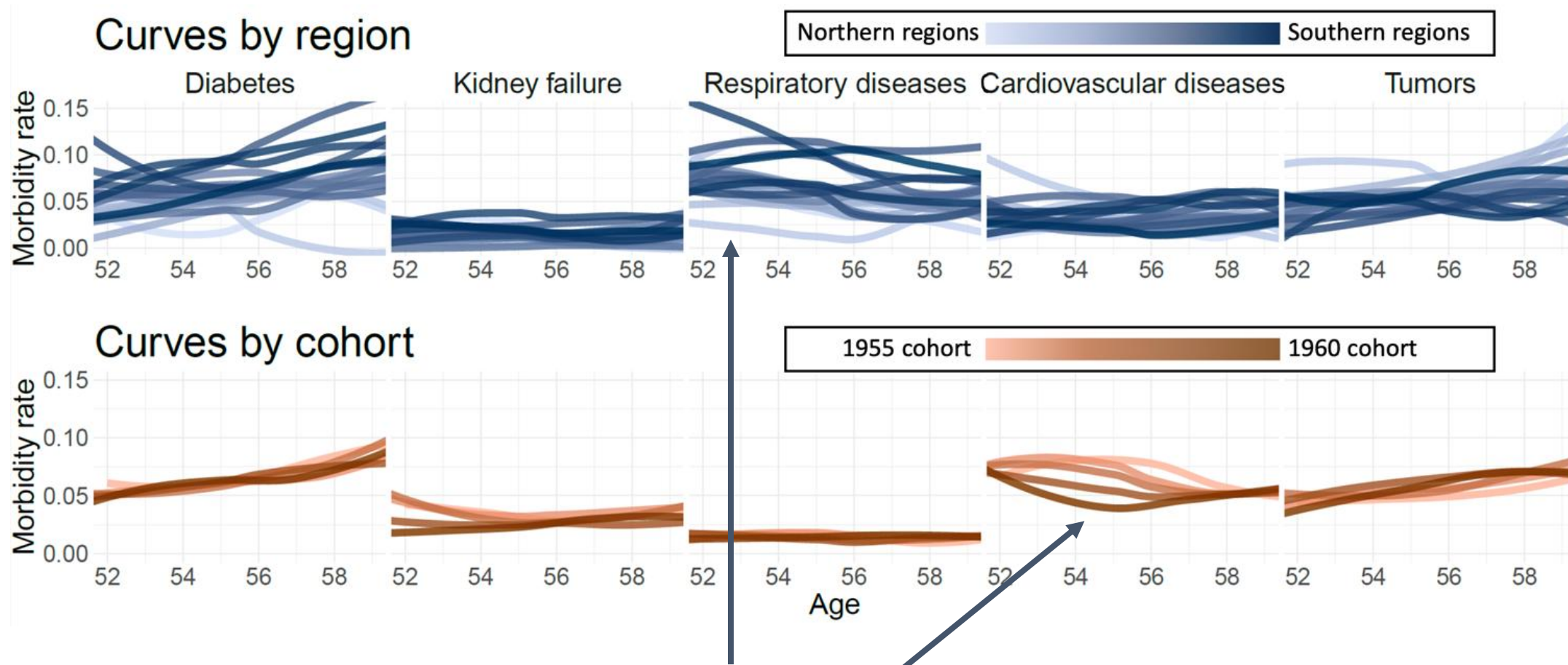
Lorenzo Schiavon



Alberto Arletti

Public health

Spatio-temporal dependence



What could explain this variability?

Multivariable logistic model

Per un soggetto i , modelliamo la presenza/assenza di D malattie y_{ij} ($j=1,\dots,D$) come:

$$y_{ij} \sim \text{Ber}(\pi_{ij}), \quad \pi_{ij} = \text{logit}^{-1}(\eta_{ij}),$$

Diagnosi (variabile risposta)

Diabete
Insufficienza renale
Malattie respiratorie
Malattie cardio-vascolari
Tumori

$$\eta_i = B(l_i, c_i) \mathbf{x}_i + \gamma \epsilon_i, \quad \epsilon_i \sim N(0, \sigma_\epsilon^2),$$

Fattori di rischio individuali

Sesso alla nascita
Età
Status economico (H/L)
Livello di istruzione (H/L)
Fumatore (Y/N)

Hierarchical multivariable logistic model

Includiamo effetti spazio-temporali

$$y_{ij} \sim \text{Ber}(\pi_{ij}), \quad \pi_{ij} = \text{logit}^{-1}(\eta_{ij}),$$

$$\eta_i = B(l_i, c_i) \mathbf{x}_i + \gamma \epsilon_i, \quad \epsilon_i \sim N(0, \sigma_\epsilon^2),$$

Stival, M., Schiavon, L., & Campostrini, S. (2024). A Bayesian approach to model local and temporal heterogeneity in repeated cross-sectional health surveys. *Journal of the Royal Statistical Society Series A: Statistics in Society*, qnae138.

Matrice dei coefficienti

Elemento generico è $\beta_{jh}(l_i, c_i)$

I coefficienti del soggetto i sono caratterizzati da:

- l_i luogo (ASL) nel quale vive il soggetto
- c_i coorte di nascita del soggetto

Communicating Complex Statistical Models in Public Health: From Science to Action

STATISTICIANS

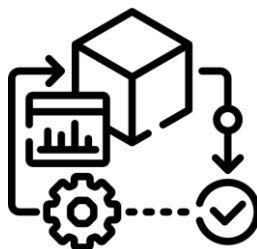


*All models are wrong,
but some are useful*

PUBLIC HEALTH STAKEHOLDERS

- POLICY MAKERS
- MD
- ASL WORKERS

Model



Understanding



**Actionable
results**

Com

So.Sta. team

SOCIOLOGISTS AND STATISTICIANS TEAM

Statistics, society, health.

So.Sta. brings together statisticians, social statisticians and sociologists working on complex social and health problem We develop and apply quantitative methods to study population health, ageing, inequalities, welfare, environmental determinants and the translation of evidence into policy.

So.Sta. stands for *Sociologists + Statisticians*. *So sta* in Venetian, *sosta* in Italian: a name about being there, stopping, and looking closer. **Stop and Think**

Meet the team

Explore our projects

So.Sta.

OBSERVE

UNDERSTAND

ACT



Interdisciplinary methods for
complex social and health
questions.

The FARSI approach

Costruire uno strumento di comunicazione...

Fast

The term refers to *all infrastructural aspects* that guarantee the **efficiency of use** of a tool, ensuring **quick processing, intuitive access to results**, and minimal computational delays.

Accessible

The term refers to a multifaceted requirement that ensures the tool and its related information can be **effectively used and understood by all potential users**.

Reliable

The term refers to the consistency, accuracy, and **trustworthiness of data, methods, and results**, ensuring they are validated, reproducible, and derived from credible sources.

Secure

The term refers to the **protection of data, privacy, and integrity** by preventing unauthorized access, ensuring compliance, and mitigating misuse.

Informative

The term refers to the ability of a tool to provide clear, relevant, and comprehensive **insights that enhance users' understanding and support informed decision-making**

The FARSI approach

Costruire uno strumento di comunicazione...

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Informative

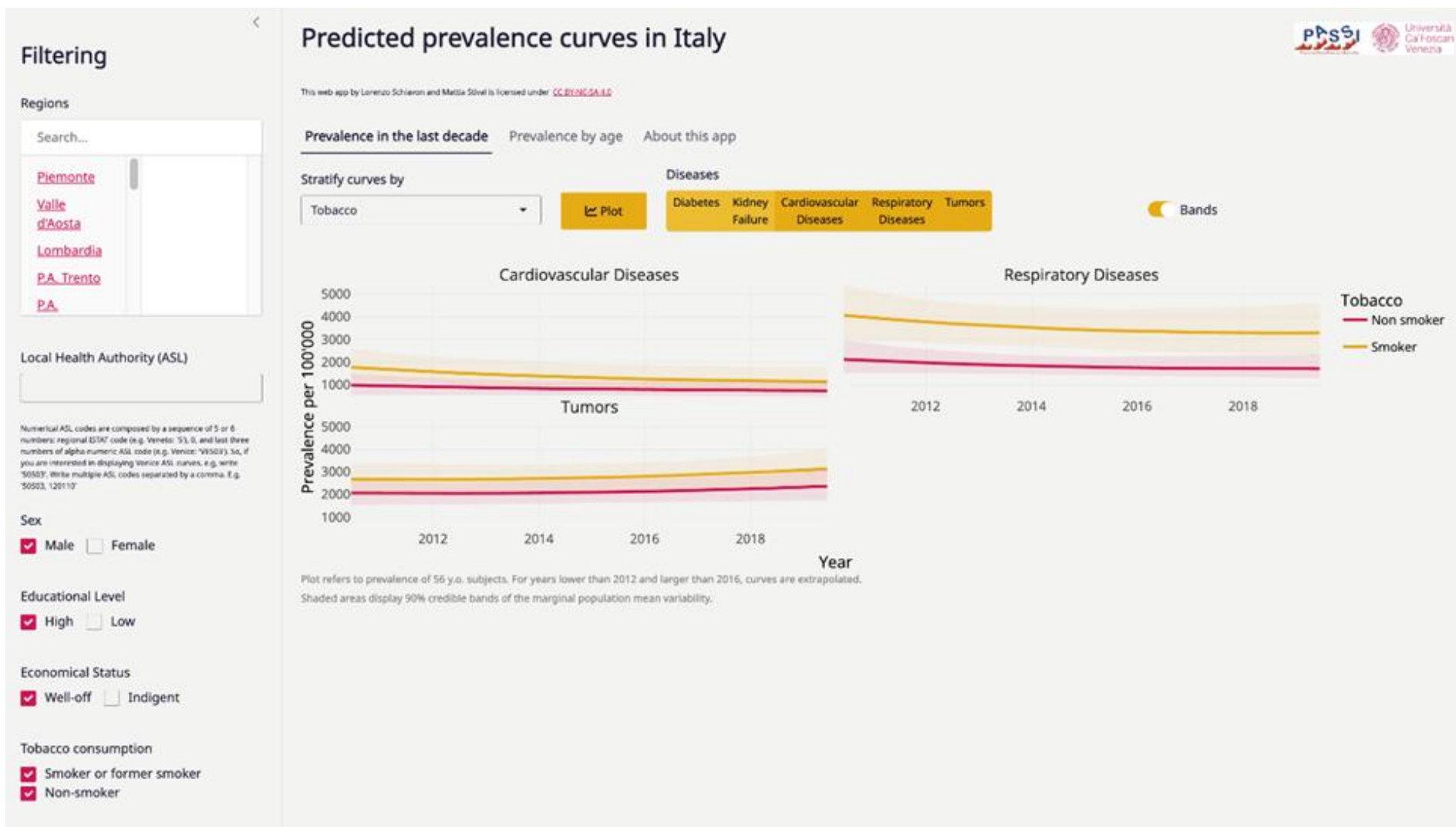
The term refers to the ability of a tool to provide clear, relevant, and comprehensive **insights that enhance users' understanding and support informed decision-making**

In pratica, abbiamo costruito una **web-app**:
<https://sosta-apps.shinyapps.io/prevalence-curves/>



Alcuni fenomeni a livello Nazionale mostrati dalla web-app

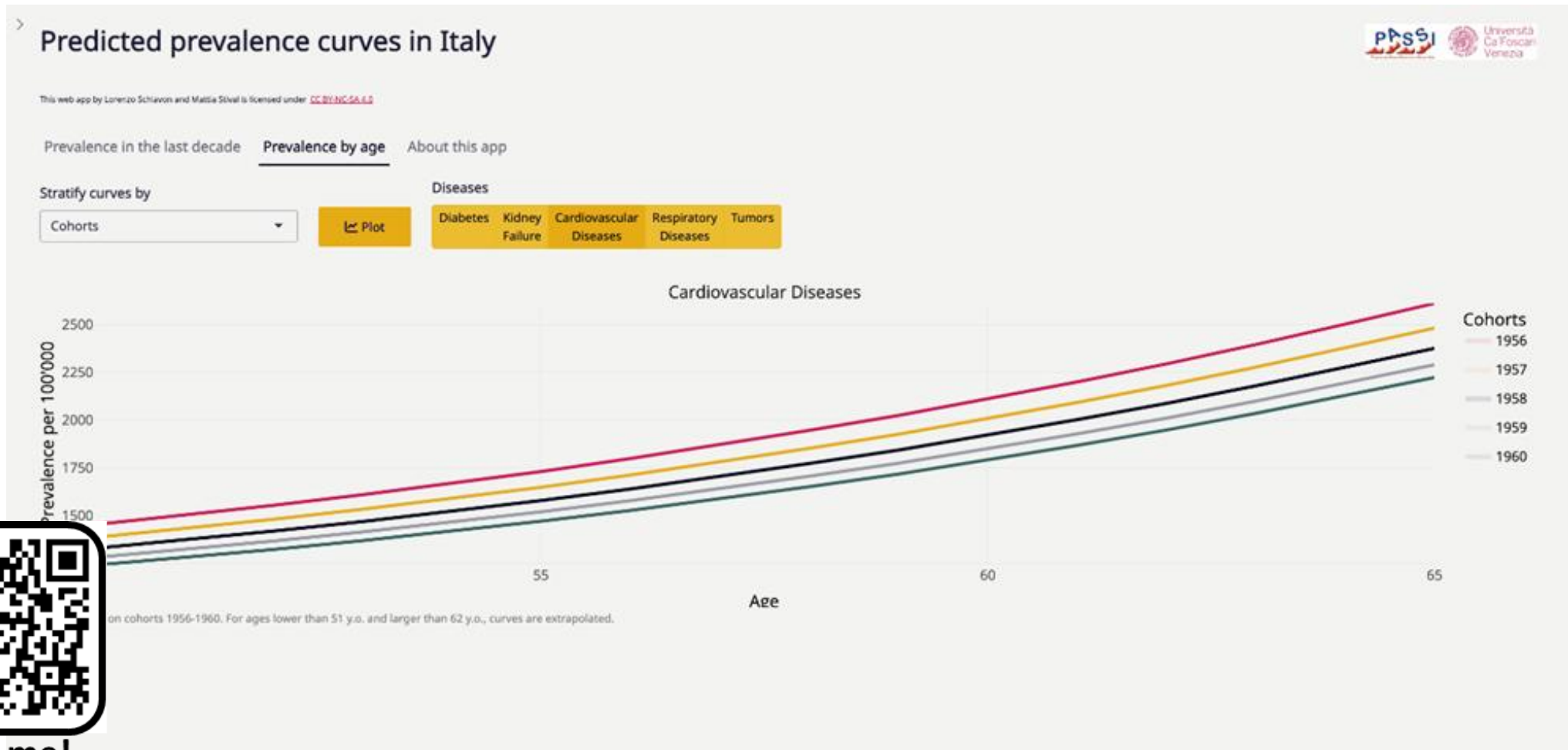
Tobacco
consumption
effect



Scan me!

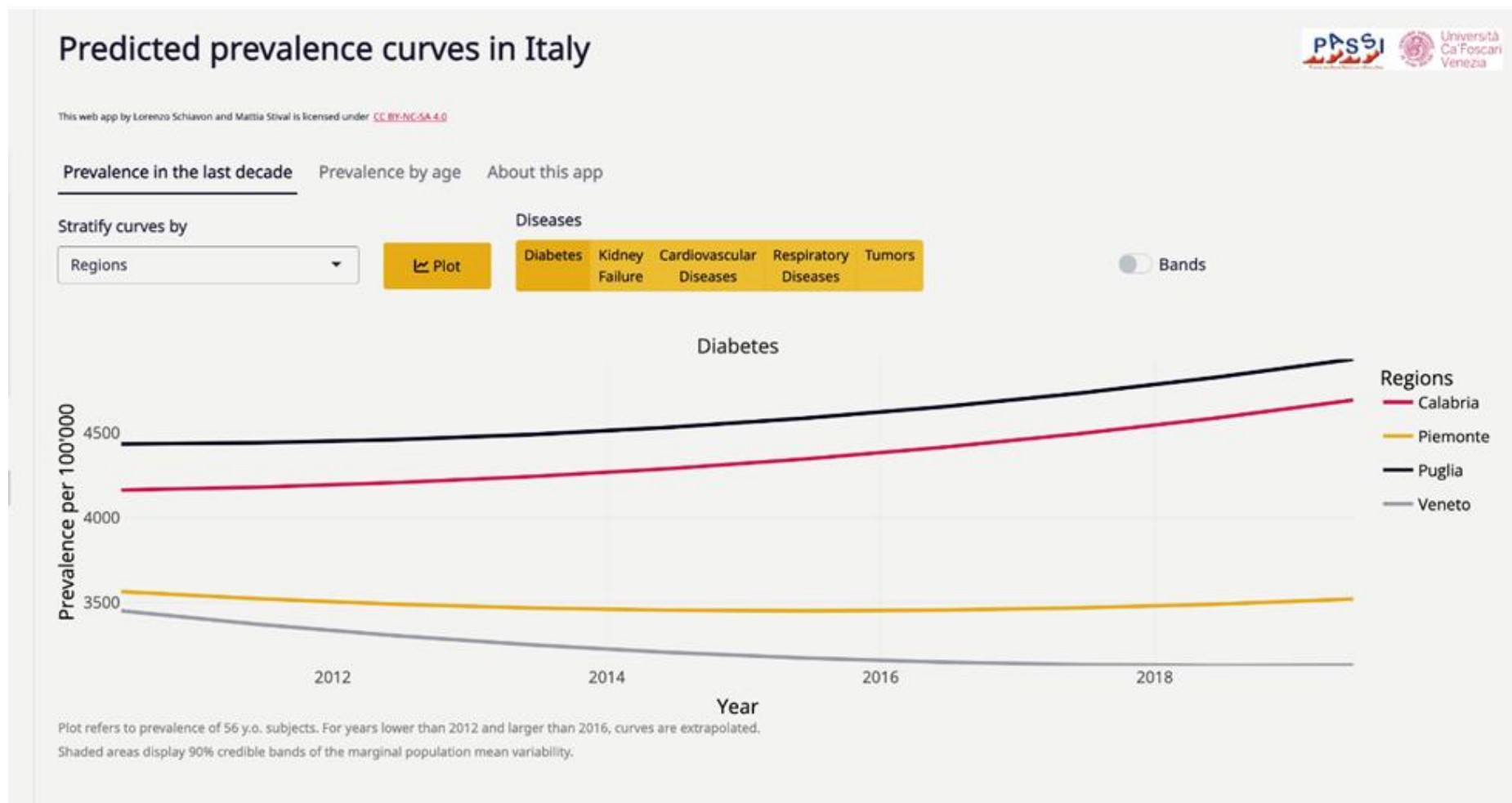
Alcuni fenomeni a livello Nazionale mostrati dalla web-app

Morbidity compression for cardiovascular diseases



Alcuni fenomeni a livello Locale mostrati dalla web-app

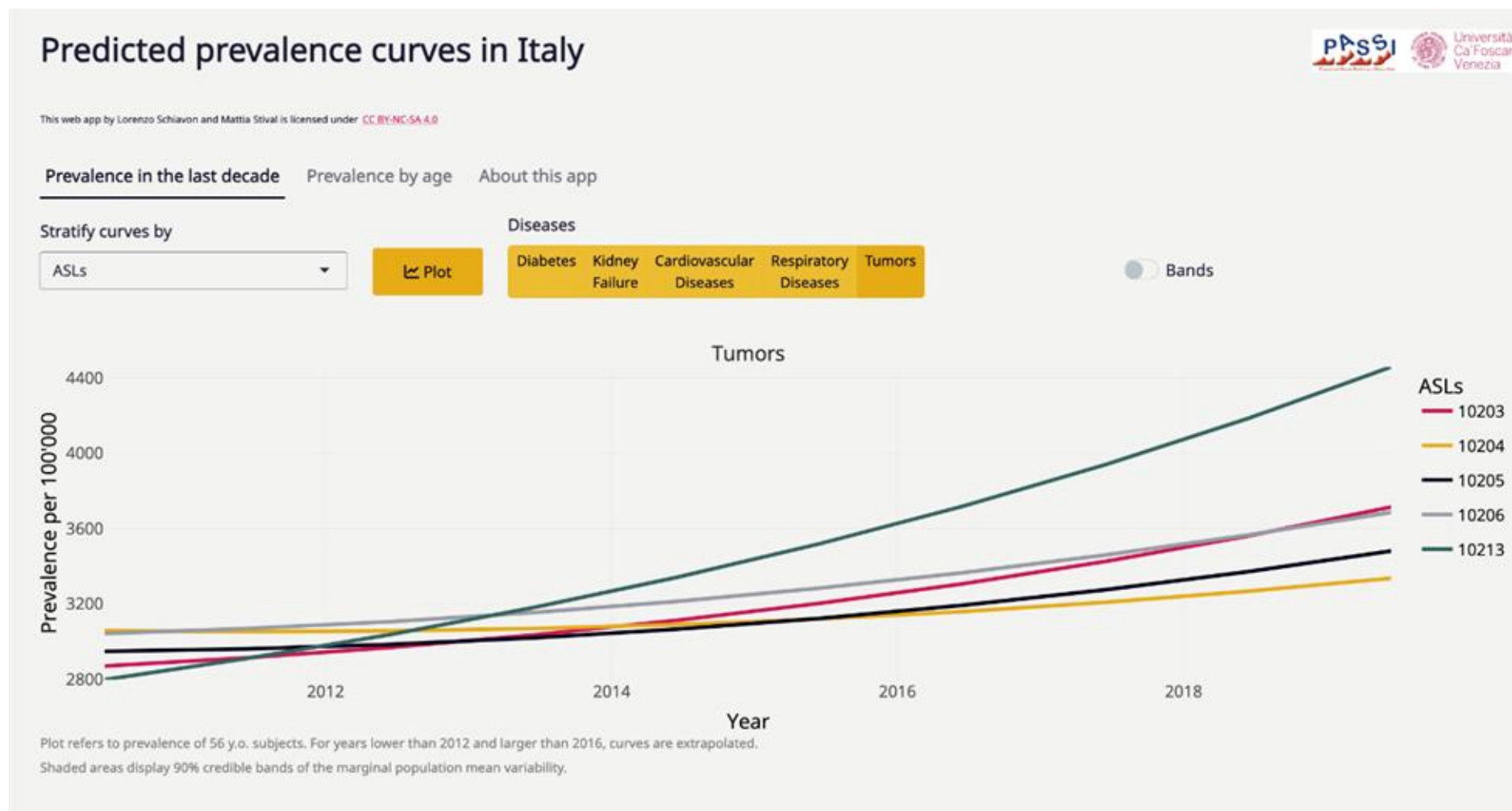
North-South inequalities in diabetes are diverging



Scan me!

Alcuni fenomeni a livello Locale mostrati dalla web-app

Asbestos effect on tumors in Alessandria's ASL



Scan me!

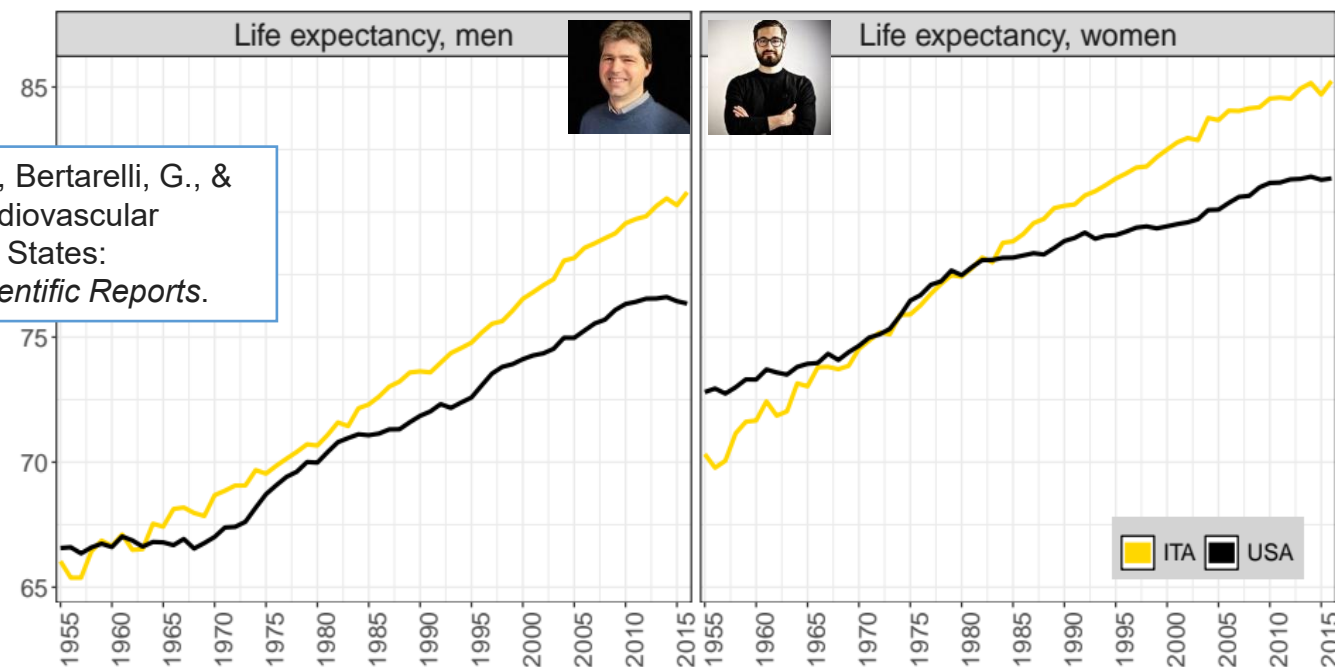
From Compression to Expansion?

Policy Recommendations from Health Surveillance Trends in Cardiovascular and Diabetes Morbidity in the US

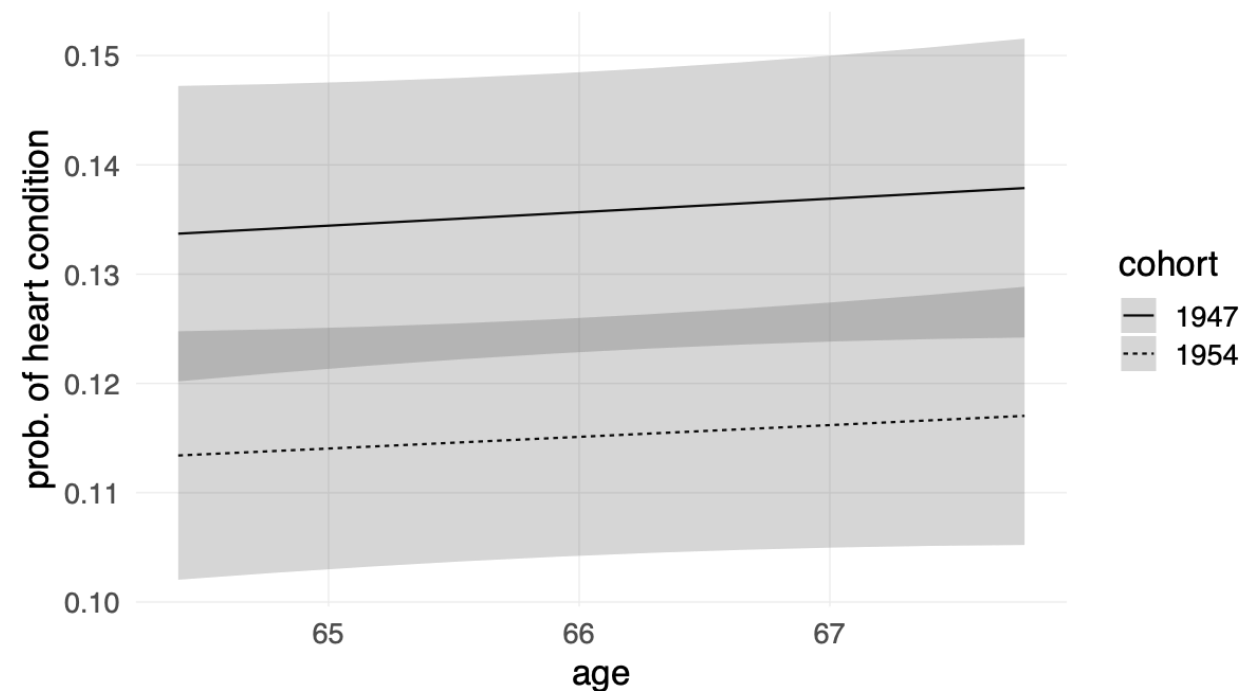
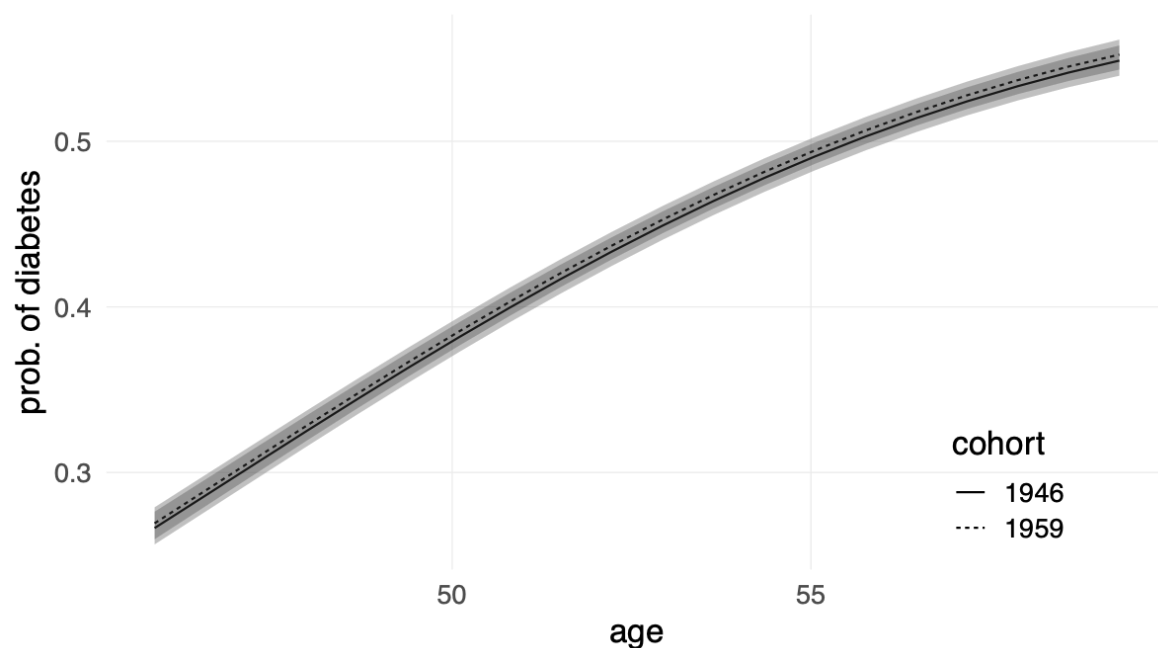


Arletti, A., Schiavon, L., Stival, M., Marzulli, M., Bertarelli, G., & Campostrini, S. (2026). Temporal trends in cardiovascular disease and diabetes prevalence in the United States: assessing compression versus expansion. *Scientific Reports*.

Aspettativa di vita in Italia e Stati Uniti
Source: Human Mortality Database



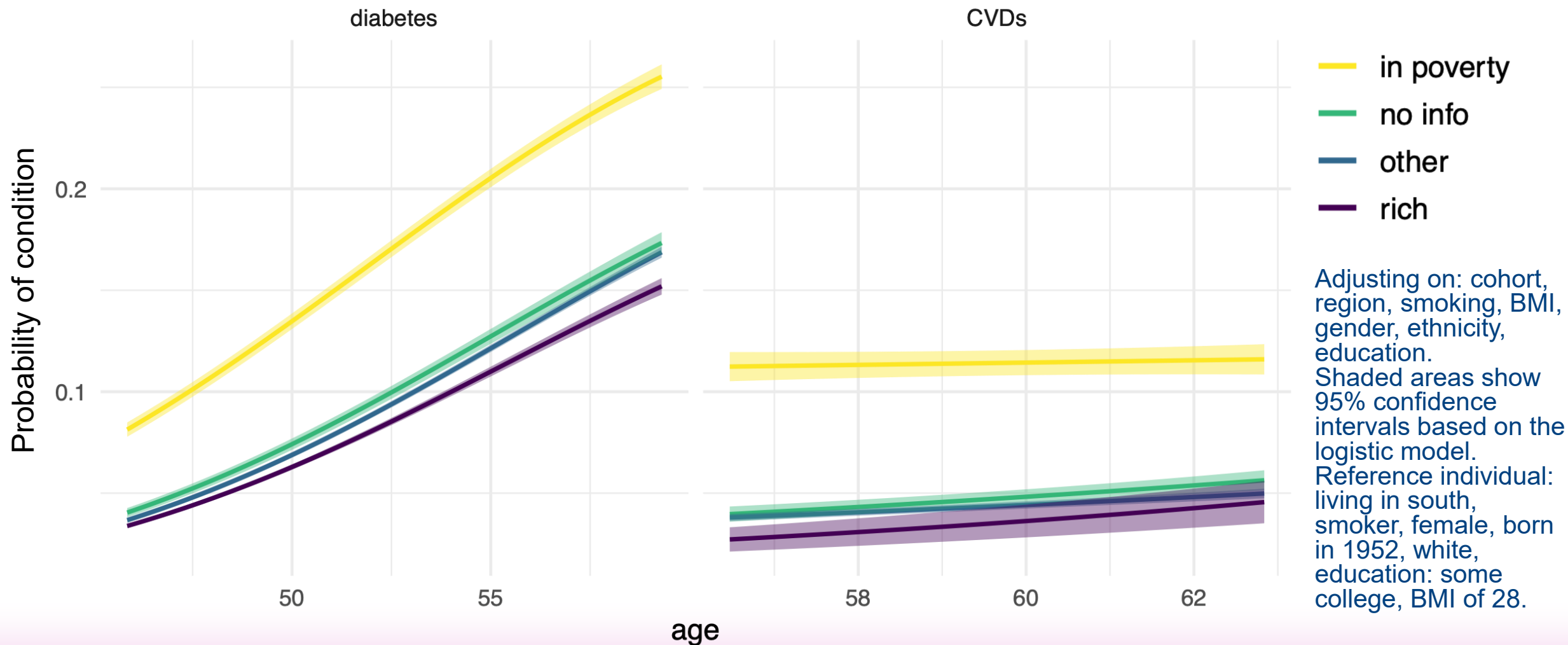
Probability of diabetes and cardiovascular diseases (CVDs) in the US aging population depending on **cohort**



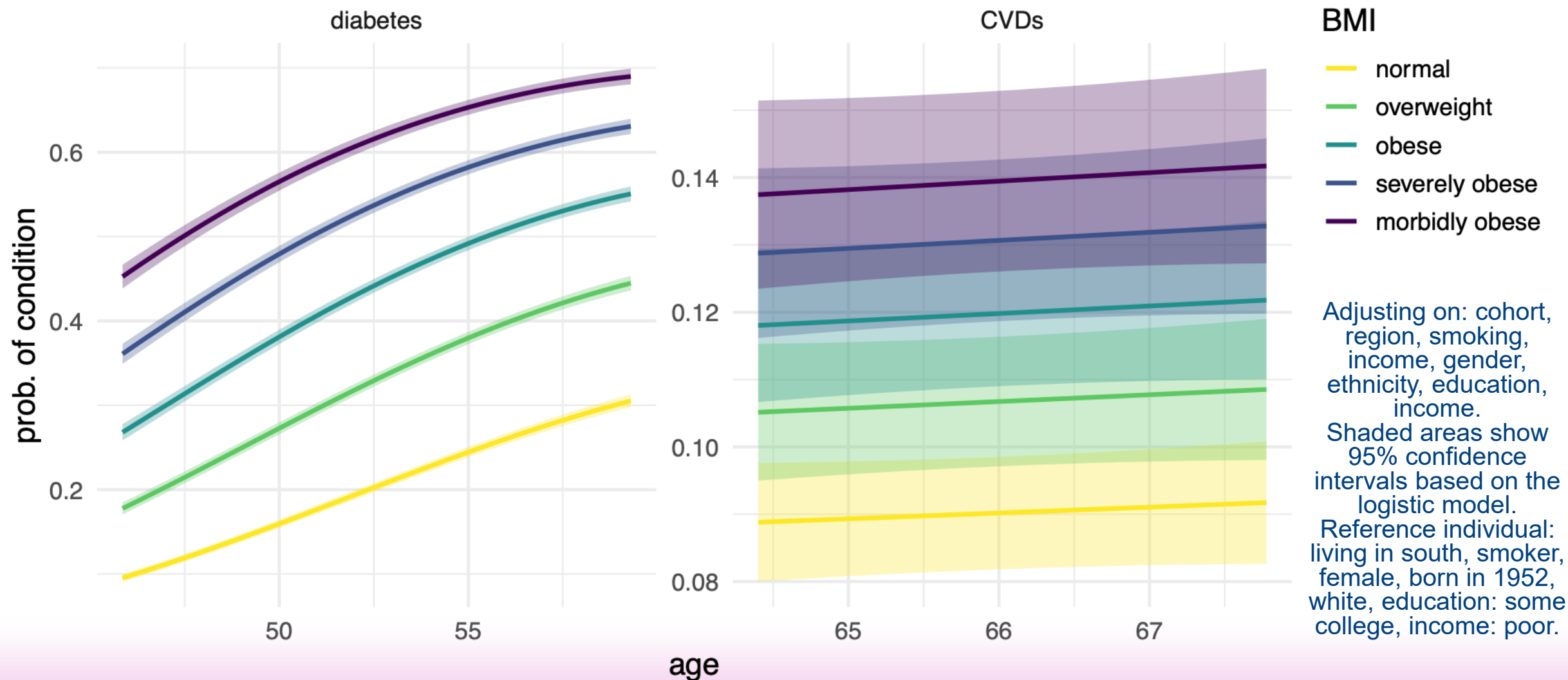
Adjusting on: region, smoking, income, gender, ethnicity, education, income, BMI.

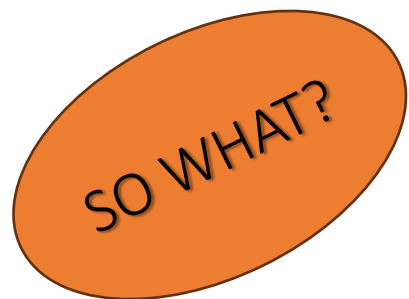
Shaded areas show 95% confidence intervals based on the logistic model. Reference individual: living in south, smoker, female, born in 1952, white, education: some college, income: poor, BMI of 28.

Probability of diabetes and cardiovascular diseases in the US aging population depending on **income**



Probability of diabetes and cardiovascular diseases in the US aging population depending on BMI





In Brief

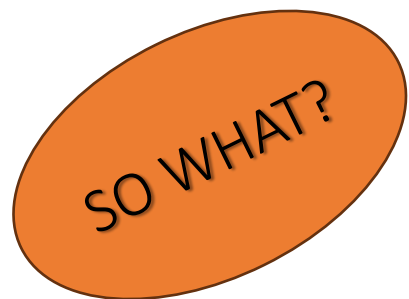
We explored **healthy aging** by examining and analyzing the **compression of morbidity**. A detailed analysis by **cohort, geographical area, and population subgroup** can reveal potential **inequalities** and provide a better understanding of the **underlying process**.

The policy implications are clear:

Risk factors matter: fostering the compression of morbidity—which **cannot be taken for granted**—requires strong support for **prevention and health promotion**.

Targeted interventions are needed to promote health and **reduce inequalities**.

Active aging matters: its role should be strengthened, and the potential impact of related interventions should be carefully assessed.



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About the statistical models:

Levium metallorum fructus in summo est; illa opulentissima sunt, quorum in alto latet vena, **adsidue plenius responsura fodienti**. (Seneca, 63 d.c., *Epistulae morales ad Lucilium*, 23, 5–6)

The research work (*fodere*): time & persons